

CSCI 3329 - Homework 3 Summary

Dataset: - File: optical_interconnection_network.csv (semicolon-separated, decimal comma) - Samples: 640 - Target (classes): Spatial Distribution (UN, HR, BR, PS)

Preprocessing: - Dropped fully-empty columns from
trailing separators - Dropped rows with missing values -
Encoded categorical columns with LabelEncoder - Scaled
features with StandardScaler

Part 2 (RepeatedKFold 10x100, accuracy): - Compared Perceptron,
LogisticRegression, KNN, GaussianNB, MLPClassifier - Best model: Neural Network
(mean=0.7083, std=0.0546)

Part 3 (Feature Selection): - Used SequentialFeatureSelector (forward) with smaller CV for
the search (10x10) - Re-evaluated selected features using the same 10x100 protocol -
Selected subsets were typically 4 features; accuracy generally decreased